Rewarding Learning

ADVANCED
General Certificate of Education
2026

Centre Number

Candidate Number

# Digital Technology

## Assessment Unit A2 1
*assessing*
## Information Systems

*ADT11*

## [ADT11]
## FRIDAY 5 JUNE, MORNING

**TIME**

2 hour 30 minutes.

**INSTRUCTIONS TO CANDIDATES**

Write your Centre Number and Candidate Number in the spaces provided at the top of this page.
**You must answer the questions in the spaces provided.**
**Do not write outside the boxed area on each page or on blank pages.**
**You are not permitted to use any calculating aid in this paper.**
Complete in black ink only. **Do not write with a gel pen.**
Answer **all six** questions.

**INFORMATION FOR CANDIDATES**

The total mark for this paper is 150.
Figures in brackets printed down the right-hand side of pages indicate the marks awarded to each question or part question.
Quality of written communication will be assessed in Questions **1(d)**, **4(c) and 5(c)**.

15026

**1** **(a)** Describe the purpose of each of the following network resources.

IP address

Wireless Access Point (WAP)

Switched hub

[9]

**(b)** Computer networks use different types of server.

Identify two different types of server and briefly describe the purpose of each.

Server

Purpose

Server

Purpose

[6]

**[Turn over**

**(c)** Describe how data is transmitted on a ring network.

[4]

**(d)** Compare peer-to-peer networks and client-server networks with regard to how network resources are shared.

**Quality of written communication will be assessed in this question.**

[6]

**[Turn over**

**2** Protocols enable networked devices to communicate with one another.

**(a)** A communication protocol will define how a message is formatted. Identify **four** other elements defined by a communication protocol.

1. _____

2. _____

3. _____

4. _____ [4]

**(b)** The Open Systems Interconnection Model (OSI) is a universal language for network communication.

**(i)** Complete the following description of the OSI model.

The OSI model consists of _____ sets and _____ layers. [2]

**(ii)** Explain the purpose of each of the following OSI layers.

Presentation

_____

_____

_____

_____

_____

_____

Data Link

_____

_____

_____

_____

_____

_____ [6]

**[Turn over**

**(iii)** Explain why the OSI model is described as providing 'a framework' for data communication on a network.

[3]

**(iv)** Describe two advantages of the OSI model.

1.

2.

[4]

**(v)** Describe the main features of each of the following communication protocols.

TCP/IP

Bluetooth

[6]

**[Turn over**

**3** A university uses the table below to record details about its courses.

| courseID | courseTitle | degree | moduleID | moduleTitle | tutorID | tutorName |
|---|---|---|---|---|---|---|
| C303 | Applied IT | BA | M101 | Databases | T2 | T Smith |
| C303 | Applied IT | BA | M102 | Business Admin | T3 | L Jones |
| C303 | Applied IT | BA | M103 | Data Analysis | T1 | R Derby |
| C304 | Data Mining | BSc | M103 | Data Analysis | T2 | T Smith |
| C304 | Data Mining | BSc | M104 | AI Development | T1 | R Derby |
| C304 | Data Mining | BSc | M105 | Data Mining | T5 | J Brown |
| C305 | Web Design | BA | M106 | HTML & CSS | T9 | S Spence |
| C305 | Web Design | BA | M103 | Data Analysis | T2 | T Smith |
| C305 | Web Design | BA | M102 | Business Admin | T2 | T Smith |

**(a)** By referring to this table, explain the difference between an entity and an attribute.

[4]

**(b)** The university wishes to store the data from the table in a normalised relational database.

**(i)** Normalise the data in the table to 1NF.

[2]

**(ii)** Normalise the data in the table to 2NF.

[3]

**(iii)** Normalise the data in the table to 3NF.

[4]

**[Turn over**

**(c)** Describe two disadvantages of normalising data in a database.

1. _____

_____

_____

_____

2. _____

_____

_____

_____ [4]

**(d)** In the box below, produce an entity-relationship (ER) model for the data in the table. It should not contain any many-to-many relationships.

[8]

**[Turn over**





**4** **(a)** During the development of a relational database, the documentation produced includes a logical data model and a physical data model.

**(i)** Explain the difference between a logical data model and a physical data model.

[4]

**(ii)** The documentation also includes a data dictionary. Identify six components of a typical data dictionary.

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____ [6]

**(b)** A college records the following details about applicants who wish to receive information on specific courses.

| studentID | surname | firstname | courseCode | email |
|---|---|---|---|---|
| S9874 | Jackson | Jane | C304 | jjackson@emailaddress.com |
| S9876 | Hall | Harry | C304 | hhall@emailaddress.com |
| S9965 | Matthews | Mark | C305 | mmatthews@emailaddress.com |
| S9984 | Hall | Anna | C304 | ahall@emailaddress.com |
| S1265 | Lyle | Lou | C303 | llyle@emailaddress.com |

**(i)** Complete the two blank lines in the following SQL statement which creates a blank database table called **studentTBL** which will store the data shown above.

```
CREATE TABLE studentTBL (

studentID varchar(5),
surname varchar (25),

_____ ,

_____ ,

email varchar(50)
);
```

[2]

**(ii)** Write the SQL statement which adds the following student's data to the table:

Kevin Knox, student ID **S2222**, email address **kknox@emailaddress.com**. This student has requested information on course **C303**.

[3]

**(iii)** The course code for Harry Hall should read **C303**, not **C304**. Write the SQL statement to update this record in the database table.

[4]

**(c)** Evaluate the use of QBE as an alternative to SQL for creating database queries.

**Quality of written communication will be assessed in this question.**

[6]

**[Turn over**

**5** **(a)** Describe how the Turing Test can determine if a machine is demonstrating human intelligence.

[4]

**(b)** An expert system is an application of artificial intelligence.

**(i)** Describe the main features of each of the following expert system components.

Shell

Fuzzy logic

_____

_____

_____

_____

_____

_____

User interface

_____

_____

_____

_____

_____

_____ [9]

**(ii)** Consider the following list of terms related to an expert system.

**Conclusions** **Inference engine** **Inferences**

**Knowledge base** **Rules** **Subject**

Use these terms to fill in the blanks in the following statement about an expert system.

The _____ interrogates the

_____ and draws _____

and _____ based on the

_____ which are stored there about the

_____ domain. [6]

**[Turn over**

**(c)** Mobile phones enable calls to be made to other mobile phones and to landlines.

Using the following terms, explain how a person travelling between two cities can make a call using a mobile phone to someone who is using a landline.

- Mobile Phone Mast
- Cells
- Handoff
- Base Station Controller
- Mobile Switching Centre
- Public Switched Telephone Network

**Quality of written communication will be assessed in this question.**

[6]

**BLANK PAGE**

**DO NOT WRITE ON THIS PAGE**

**(Questions continue overleaf)**

**[Turn over**

**6** **(a)** Data mining and big data are widely used by organisations such as banks.

**(i)** Explain the difference between the following:

Data mining

Big data

[4]

**(ii)** Describe how banks use data mining to help prevent fraudulent activity on a customer's account.

[3]

**(b)** The Data Protection Act requires that all personal data must be stored securely.

**(i)** Describe three methods used by banks to ensure customer data is kept secure.

1. _____

_____

_____

_____

2. _____

_____

_____

_____

3. _____

_____

_____

_____ [6]

**(ii)** State three other requirements of the Data Protection Act.

1. _____

_____

2. _____

_____

3. _____

_____ [3]

**[Turn over**

The Data Protection Act identifies a number of key people.

**(iii)** Describe the role of the Data Controller.

_____

_____

_____

_____

_____

_____ [3]

**(iv)** State two other key people identified in the Data Protection Act.

1. _____

_____

2. _____

_____ [2]

**(c)** Describe two steps a bank should take to make sure employees comply with the Computer Misuse Act.

1. _____

_____

_____

_____

2. _____

_____

_____

_____ [4]

**THIS IS THE END OF THE QUESTION PAPER**

**BLANK PAGE**

**DO NOT WRITE ON THIS PAGE**

**BLANK PAGE**

**DO NOT WRITE ON THIS PAGE**

**DO NOT WRITE ON THIS PAGE**

| For Examiner's use only | |
|---|---|
| **Question Number** | **Marks** |
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| **Total Marks** | |

**Examiner Number**

Permission to reproduce all copyright material has been applied for.
In some cases, efforts to contact copyright holders may have been unsuccessful and CCEA will be happy to rectify any omissions of acknowledgement in future if notified.